

Chapter 9

Commitment Schemes

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Plan

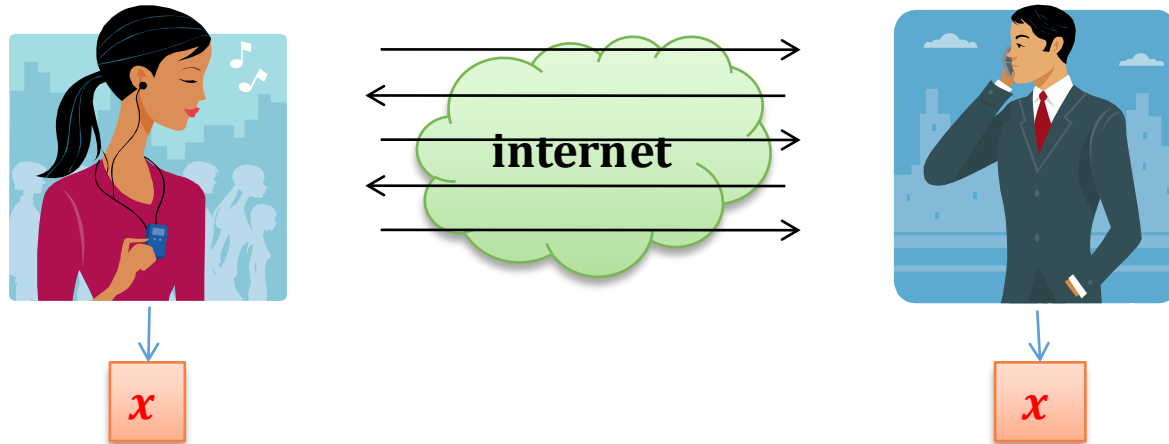


1. Coin-flipping by telephone
2. Commitment schemes

Coin-flipping by telephone [Blum'81]

privacy and authenticity is not a problem

Suppose Alice and Bob are connected by a secure internet link:



The goal of Alice and Bob is to toss a coin.

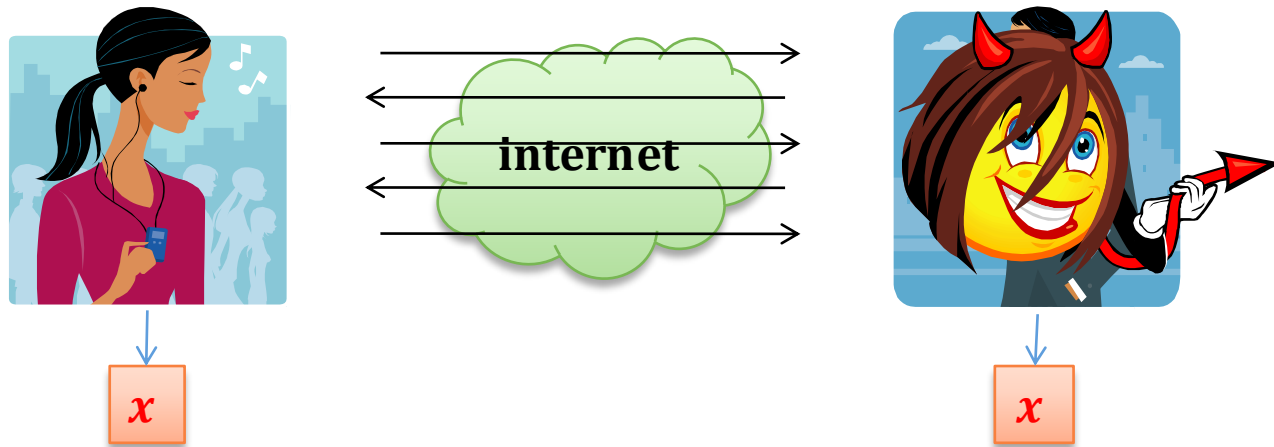
In other words:

They want to execute some protocol π in such a way that at the end of the execution they both output the same bit x distributed uniformly over $\{0, 1\}$.

How to define security? [1/2]

Let us just stay at an informal level...

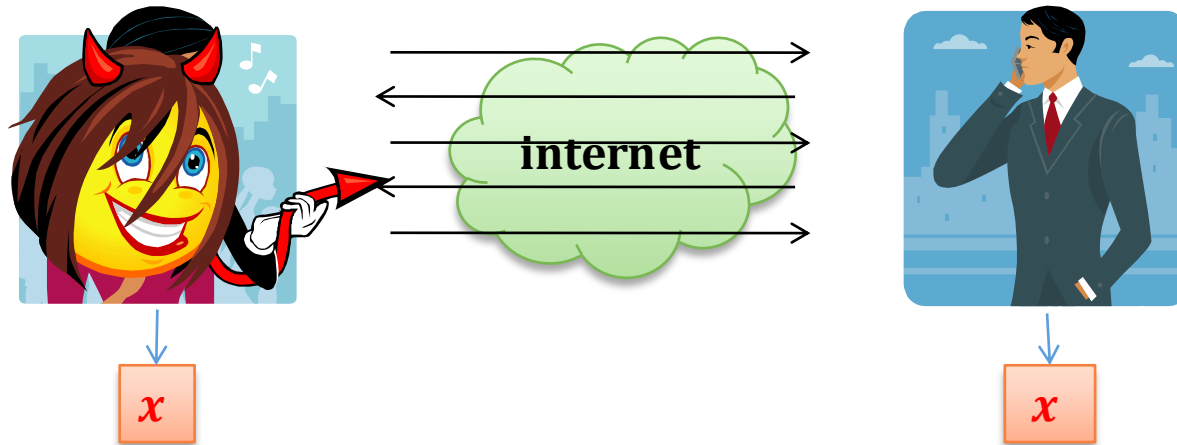
From the point of view of Alice:



even if **Bob** is **cheating** (i.e.: he doesn't follow the protocol):
if the protocol terminates successfully, then x has a
uniform distribution

How to define security? [2/2]

The same holds from the point of view of Bob



even if Alice is **cheating** (i.e.: he doesn't follow the protocol):
if the protocol terminates successfully, then **x** has a
uniform distribution

Note the difference

Unlike what we saw on the previous lectures:

the enemy can be one of the parties

(**not** an external entity)

A cheating party is sometimes called a **corrupted** party, or a **malicious** party.

We will see many other examples of this later!

How to solve this problem?

Idea

Remember the old
game:

rock-paper-scissors?





Alice

Bob

			
	draw	Alice wins	Bob wins
	Bob wins	draw	Alice wins
	Alice wins	Bob wins	draw

Let's simplify this game

		Alice	
		$A = 0$	$A = 1$
Bob	$B = 0$	Alice wins	Bob wins
	$B = 1$	Bob wins	Alice wins

In other words: Alice wins iff $A \oplus B = 0$.

Another way to look at it

Alice
has an input **A**

Bob
has an input **B**

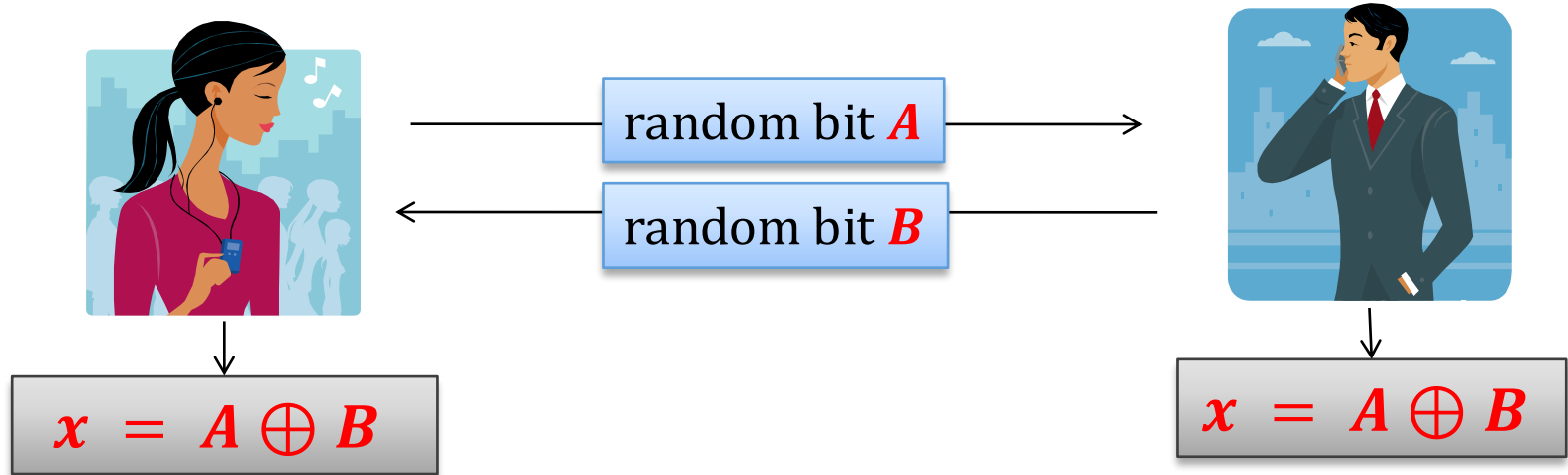


they should jointly compute

$$x = A \oplus B$$

(in a secure way)

What to do?



Problem:

A and **B** should be sent at the same time

(e.g. if **A** is sent before **B** then a malicious **Bob** can set $B := x \oplus A$, where x is chosen by him).

How to prevent this?

Seems hard:

the internet is not synchronous...

A solution:

bit commitments

Plan

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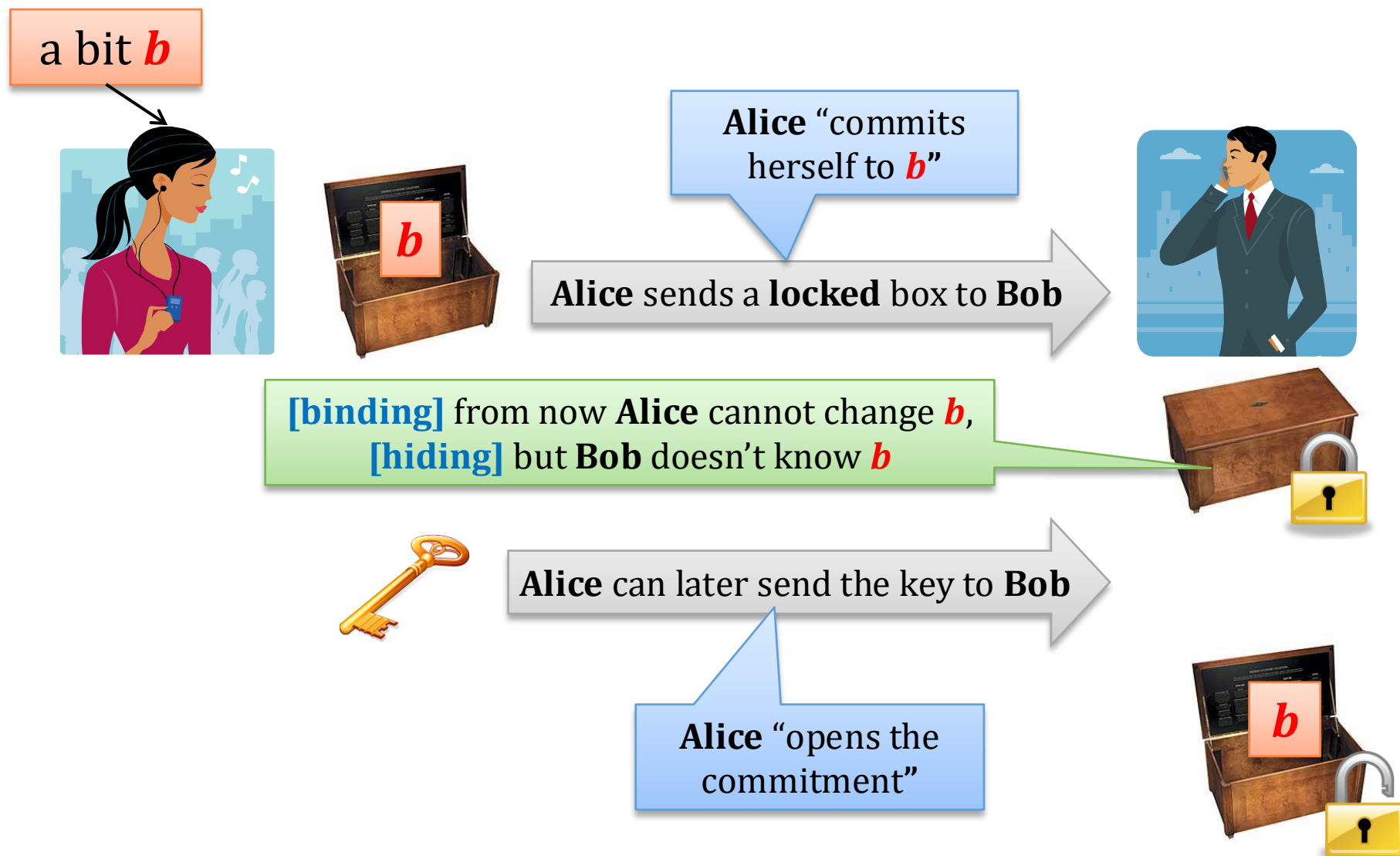
1. definition

2. construction based on QRA

3. construction based on discrete log

4. construction based on PRG

Commitment schemes – an intuition



Commitment schemes – a functional definition

A **commitment scheme** is a protocol executed between Alice and Bob consisting of two phases: **commit** and **open**.

In the **commit** phase:

- **Alice** takes some input bit **b** .
- **Bob** takes no input.

In the **open** phase:

- **Alice** outputs nothing
- **Bob** outputs **b** , or **error**

Security requirements - informally

[binding]

After the **commit** phase there exists at most one value ***b*** that can be open in the **open** phase.

[hiding]

As long as the **open** phase did not start **Bob** has no information about ***b***.

How to define security formally?

Not so trivial – remember that the parties can misbehave arbitrarily.

We do not present a complete definition here.

(The hiding property can be defined using the “**indistinguishability**” principle.)

The definition depends on some options.

1. What is the computational power of a **cheating Alice**?
2. What is the computational power of a **cheating Bob**?

The computational power of the adversary

If a cheating Alice can be infinitely powerful, we say that the protocol is **unconditionally binding**.

Otherwise it is **computationally binding**.

If a cheating Bob can be infinitely powerful, we say that the protocol is **unconditionally hiding**.

Otherwise it is **computationally hiding**.

Of course, to be formal we would need to introduce a security parameter...

Unconditionally hiding and binding commitment schemes do not exist

Proof (intuition)

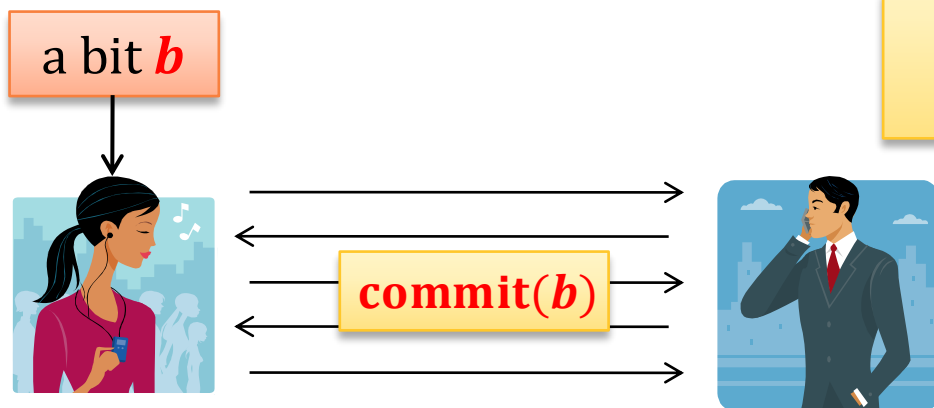
There are two options:

1. there exists a way to open **1 - b**, or

in this case “infinitely powerful”
Alice can cheat

2. there doesn't exist such a way

in this case “infinitely powerful”
Bob can learn **b**



So, how does it solve the coin-flipping problem?

chooses a random
bit $A \leftarrow \{0, 1\}$



commits to A



chooses a random
bit $B \leftarrow \{0, 1\}$

sends B

opens A

output
 $A \oplus B$

output
 $A \oplus B$

Problem

Alice can always refuse to send the last message.

This is unavoidable (there has to be the **last message** in the protocol).

But they can use a convention:
if **Alice** didn't send the last message – she lost!

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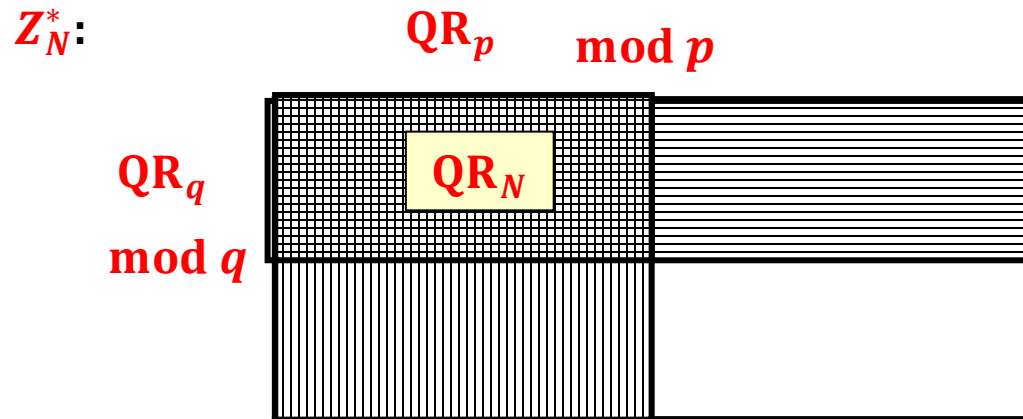
4. construction based on PRG



Remember quadratic residues modulo an RSA modulus?

$$\text{QR}_N = \{x^2 \bmod N : x \in \mathbb{Z}_N^*\}$$

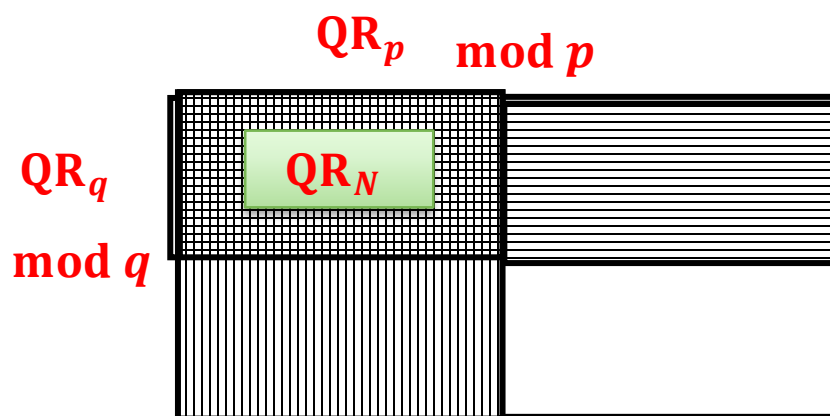
Fact: For $N = pq$ we have $|\text{QR}_N| = |\mathbb{Z}_N^*| / 4$.



Jacobi Symbol

for any prime p define $J_p(x) := \begin{cases} +1 & \text{if } x \in \mathbf{QR}_p \\ -1 & \text{otherwise} \end{cases}$

for $N = pq$ define $J_N(x) := J_p(x) \cdot J_q(x)$



$J_N(x) :=$

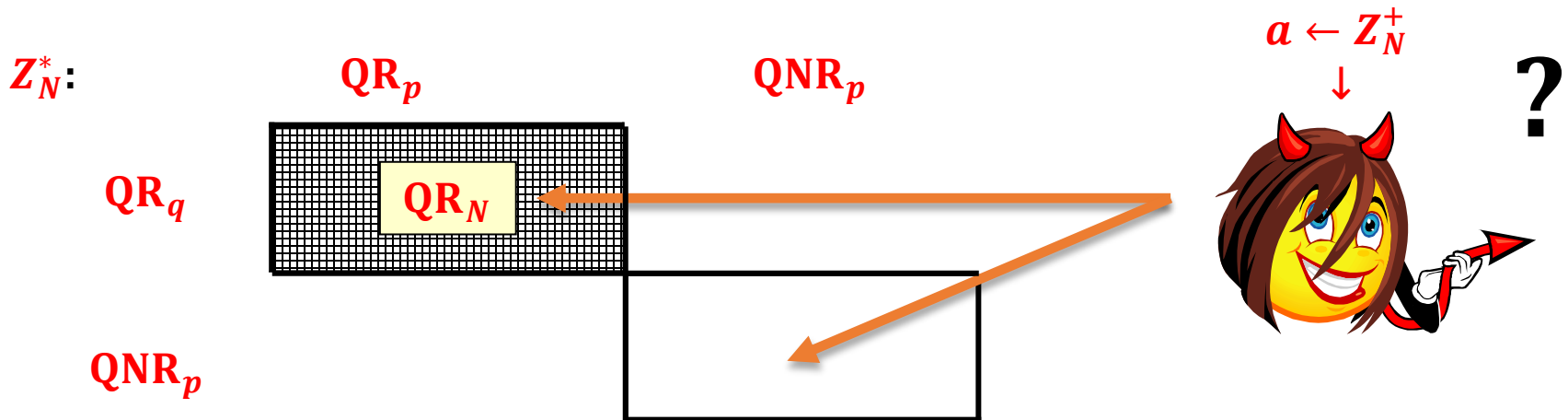
$+1$	-1
-1	$+1$

It is a subgroup of $\mathbf{Z}_N^*$

$$\mathbf{Z}_N^+ := \{x \in \mathbf{Z}_N^* : J_N(x) = +1\}$$

Jacobi symbol can be computed efficiently!
(even in p and q are unknown)

Quadratic Residuosity Assumption



Quadratic Residuosity Assumption (QRA):

For a random $a \leftarrow Z_N^+$ it is computationally hard to determine if $a \in QR_N$.

Formally: for every **polynomial-time** probabilistic algorithm D the value:

$$\left| P(D(N, a) = Q_N(a)) - \frac{1}{2} \right|$$

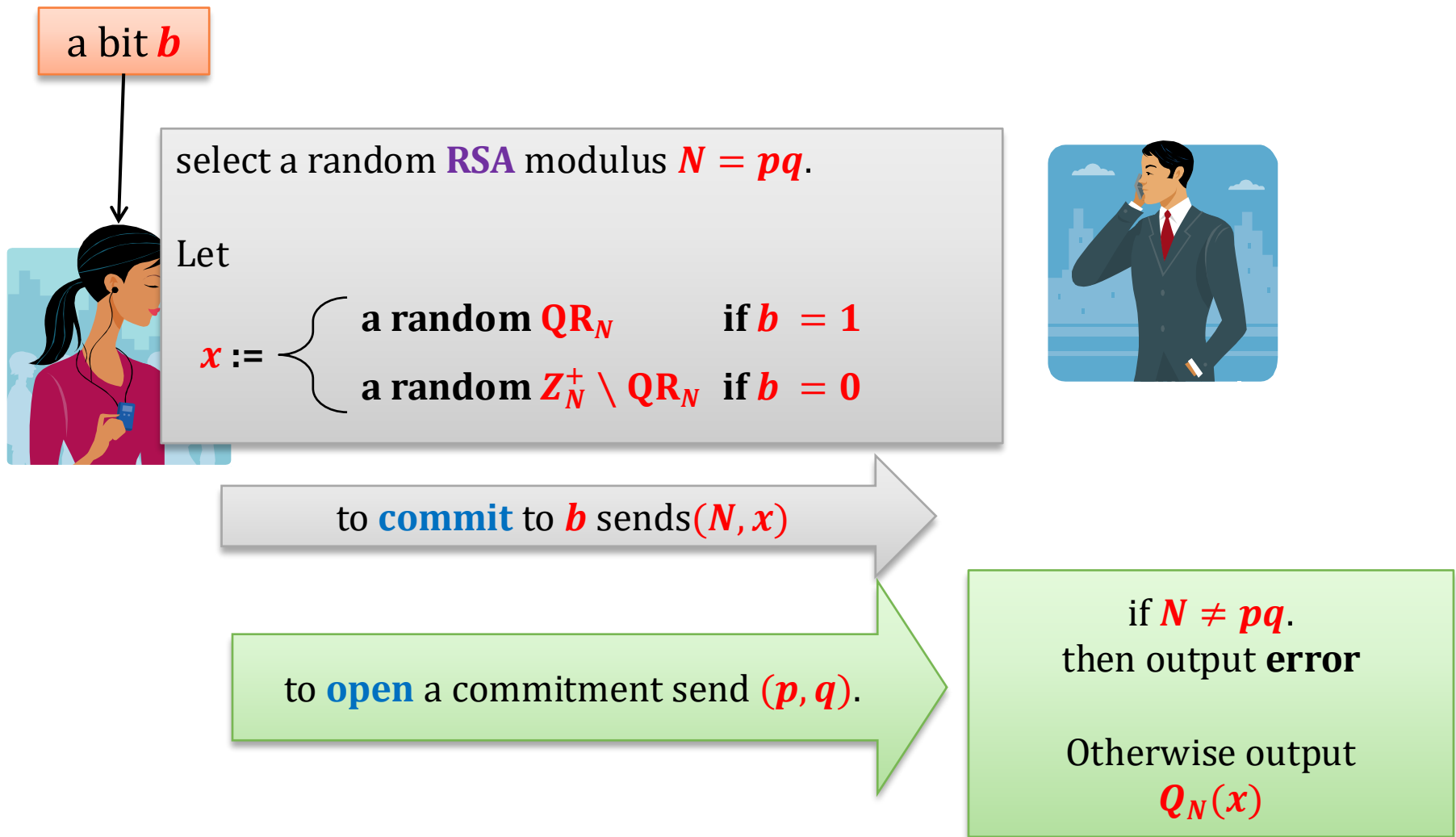
(where $a \leftarrow Z_N^+$) is **negligible**.

Where a predicate $Q_N: Z_N^+ \rightarrow \{0, 1\}$ is defined as follows:

$Q_N(a) = 1$ if $a \in QR_N$

$Q_N(a) = 0$ otherwise

A construction based on QRA



This commitment scheme is unconditionally binding

Why?

Suppose **Alice** has sent (N, x) to **Bob**.

What can **Bob** output at the end of the opening phase?

There exists the following options:

- N is not an **RSA** modulus – in this case **Bob** will always output **error**,
- $x \in \text{QR}_N$ – in this case **Bob** can only output **0** or **error**,
- $x \notin \text{QR}_N$ – in this case **Bob** can only output **1** or **error**.

This commitment scheme is computationally hiding, assuming QRA holds

Proof (intuition)

To distinguish between $b = 0$ and $b = 1$ a malicious **Bob** would need to distinguish QR_N from the other elements of Z_N^* ...

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A construction based on discrete log

G – a group in which discrete log is hard

g – a generator of G

a bit b

Select a random

$$y \leftarrow Z_{|G|}$$

Let

$$x := \begin{cases} g^y & \text{if } b = 0 \\ s \cdot g^y & \text{if } b = 1 \end{cases}$$

Select s – a random element of G

G, g, s



to commit to b send x

to open a commitment send y .

if $x = g^y$ output $b = 0$
if $x = s \cdot g^y$ output $b = 1$

This commitment scheme is **computationally binding**, assuming that the discrete log is hard in G

Proof (intuition)

To be able to open the commitment in two ways, a **cheating Alice** needs to know y and y' such that there exists x such that

$$g^y = x = s \cdot g^{y'}$$

But this means that $g^{y-y'} = s$. So, she would know the discrete log of s .

This commitment scheme is unconditionally hiding

Why?

Because x is just a random element of G .

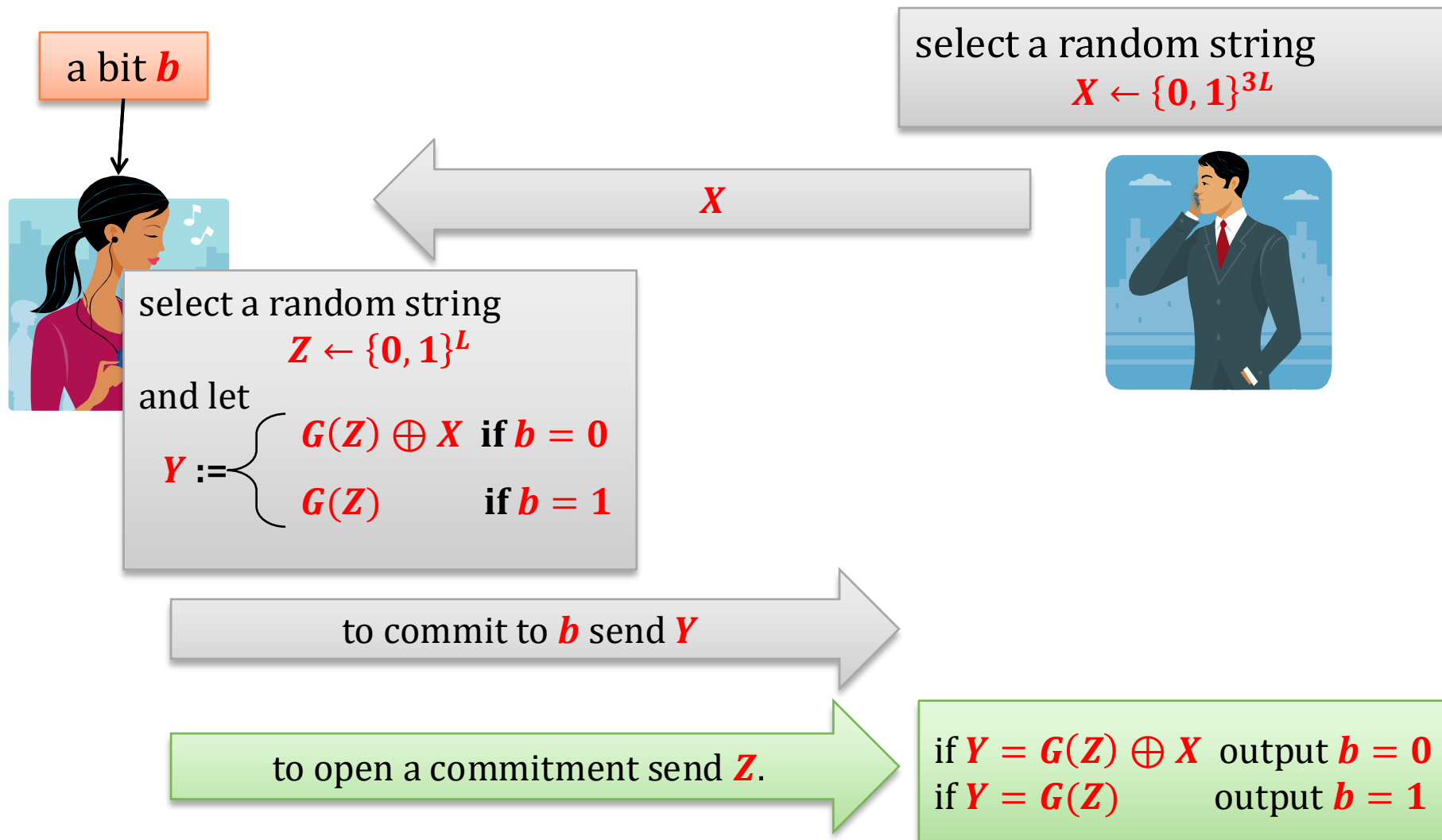
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A construction based on PRGs [Naor'91]

$G: \{0, 1\}^L \rightarrow \{0, 1\}^{3L}$ a PRG



This commitment scheme is unconditionally binding

Proof (intuition)

To be able to open the commitment in two ways, a **cheating Alice** needs to find Z and Z' such that there exists Y such that:

$$G(Z) \oplus X = Y = G(Z')$$

This means that $G(Z) \oplus G(Z') = X$.

How many X 's have the property that

there exist Z and Z' such that $G(Z) \oplus G(Z') = X$?

By the counting argument: at most $(2^L)^2 = 2^{2L}$.

Therefore, the probability that a **random $X \in \{0, 1\}^{3L}$** has this property is at most

$$\frac{2^{2L}}{2^{3L}} = 2^{-L}.$$

QED

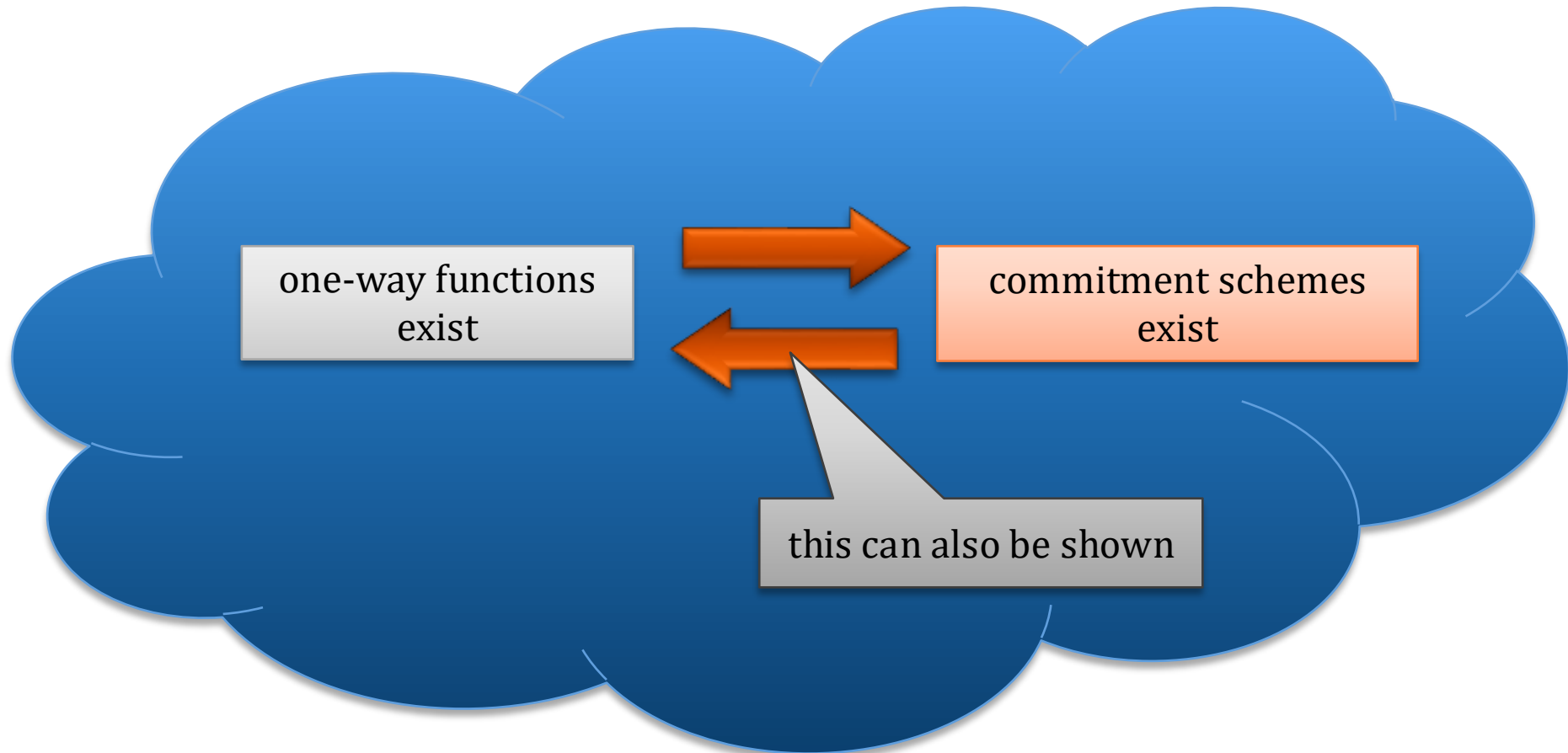
This commitment scheme is computationally hiding, assuming G is a secure PRG

Why?

Obviously, if, instead of $G(Z)$ Alice uses a completely random string R , then the scheme is secure against a cheating Bob.

If a scheme behaved differently with R and with $G(Z)$, then a cheating Bob could be used as a distinguisher for G .

Moral



Commitment schemes are a part of **Minicrypt**!

String commitment

How to commit to a longer string $\mathbf{x} = (x_1, \dots, x_n)$?

Just commit to every x_i separately.

To open the commitment, open each commitment to x_i .

(**Bob** accepts only if all the openings were **ok**)

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